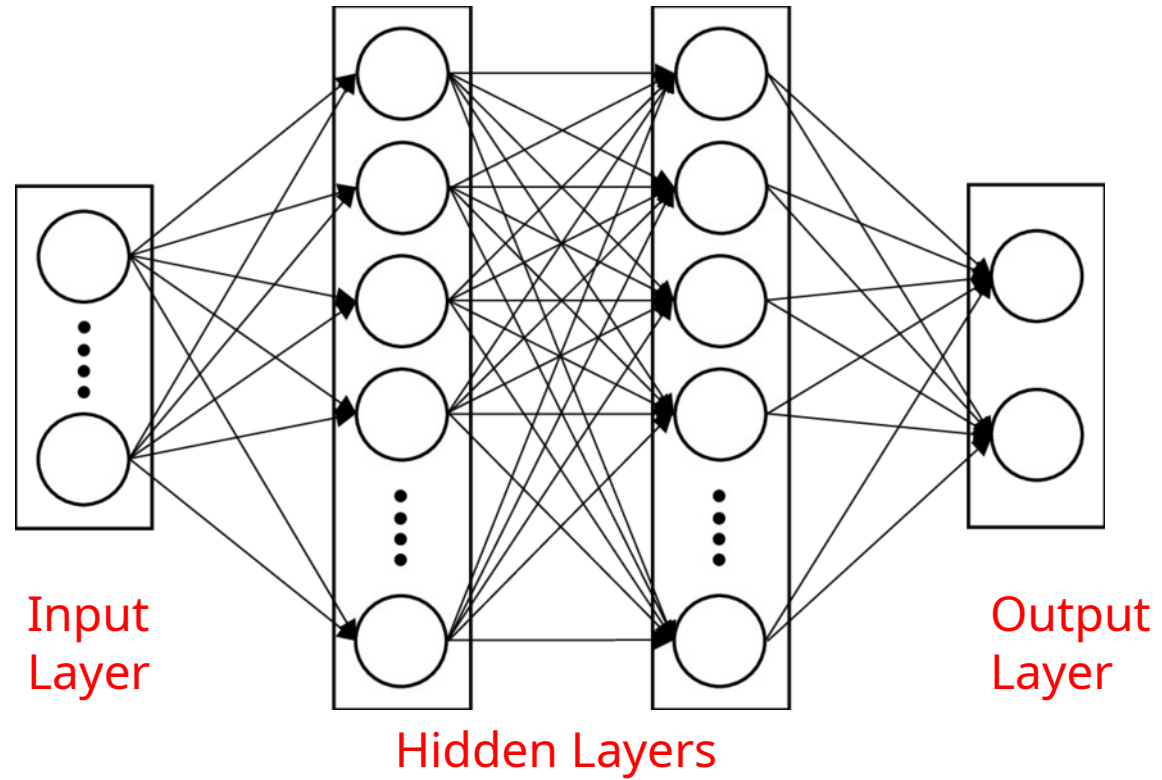


Deep Learning

Convolutional Neural Networks

Neural Network



Inductive Bias

- Set of (explicit or implicit) assumptions made by a learning algorithm in order to perform prediction
 - For example,
 - Linearity assumption in Linear Regression and Logistic Regression
 - Fully-connected structure of the neural network
 - Hierarchical Representation Bias
 - Others

Image Classification

- Core task in Computer Vision



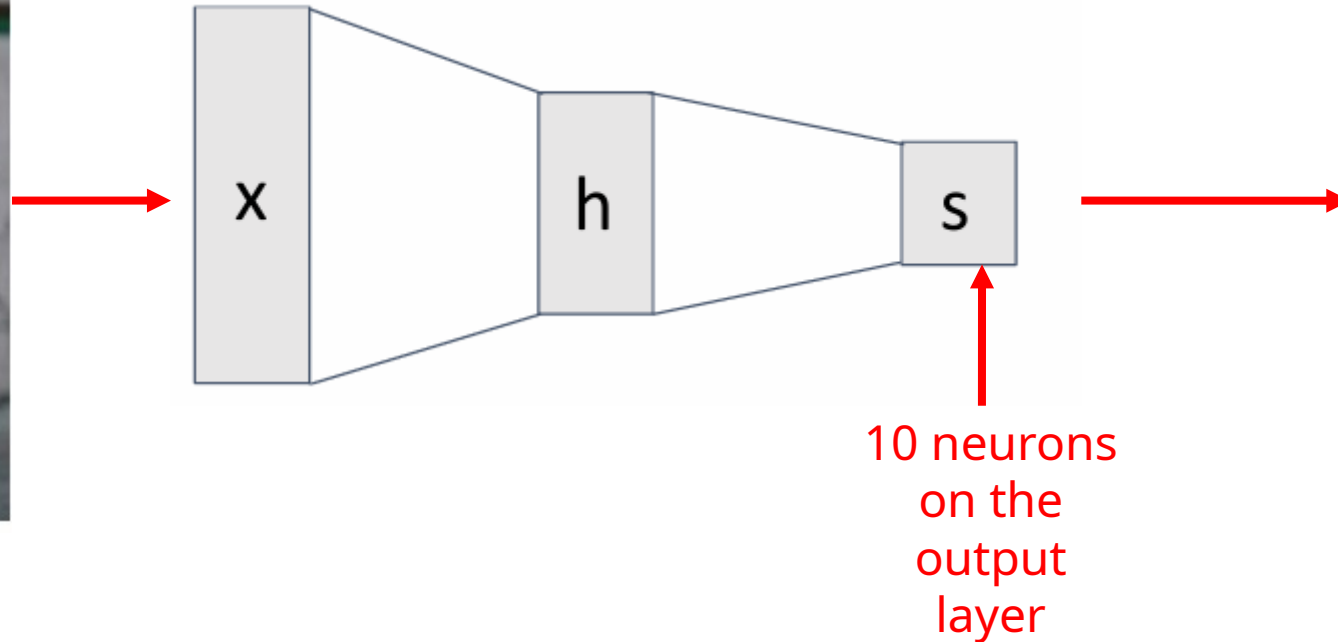
- Given an image and a set of labels

- Cat
- Dog
- Plane
- Truck

Image Classification

- Core task in Computer Vision

- Given an image and a set of 10 labels

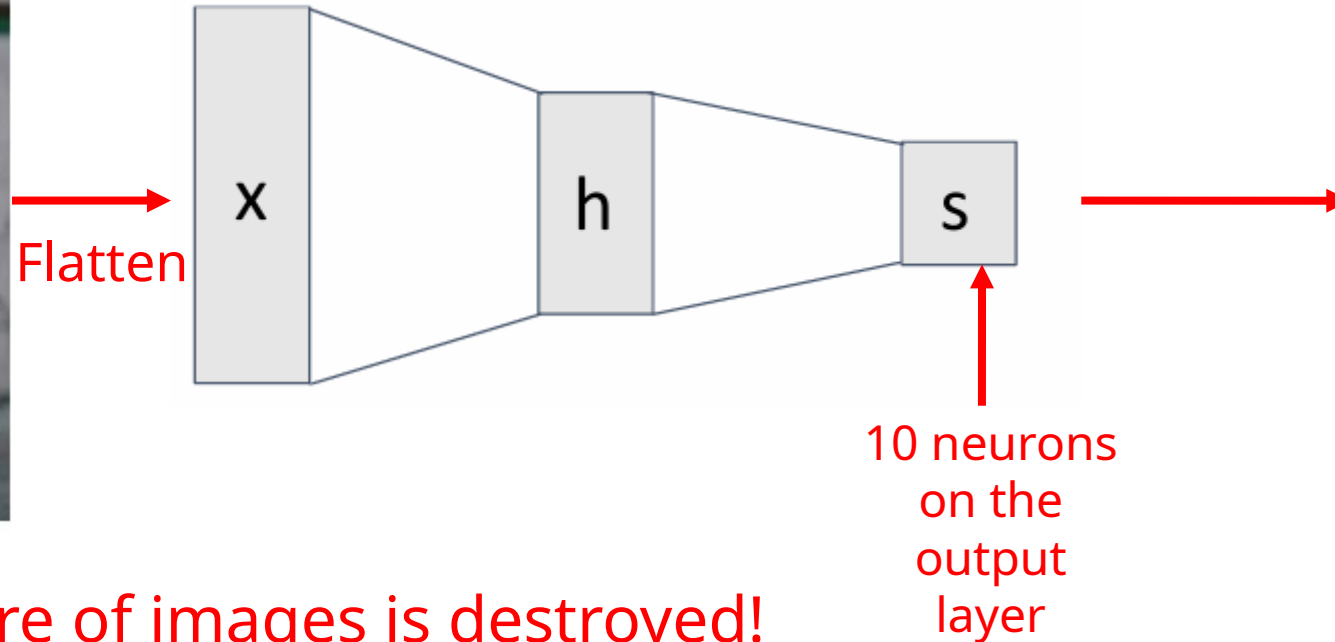


0.6	<i>Cat</i>
0.05	<i>Dog</i>
0.02	<i>Truck</i>
⋮	⋮
0.1	Plane
0.09	Duck
0.03	Horse

Image Classification

- Core task in Computer Vision

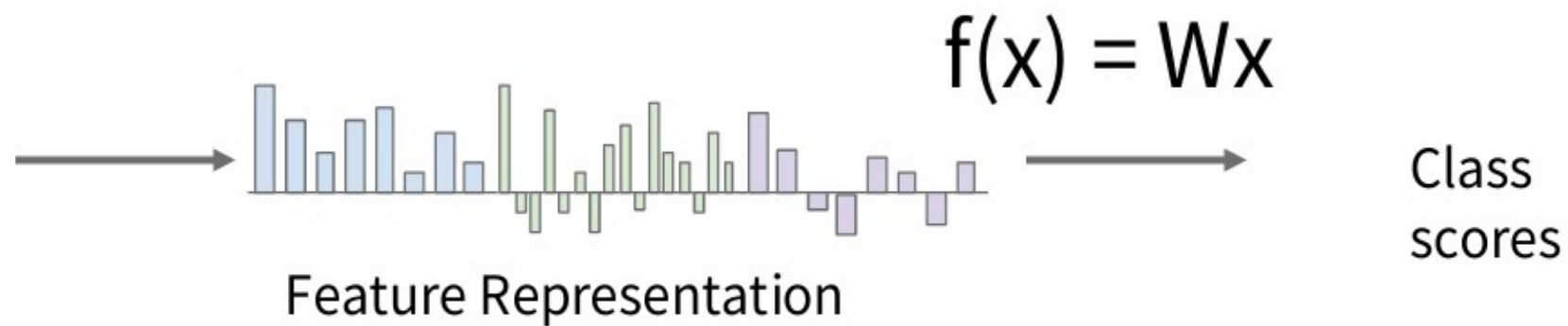
- Given a set of 10 labels



0.6	<i>Cat</i>
0.05	<i>Dog</i>
0.02	<i>Truck</i>
⋮	⋮
0.1	Plane
0.09	Duck
0.03	Horse

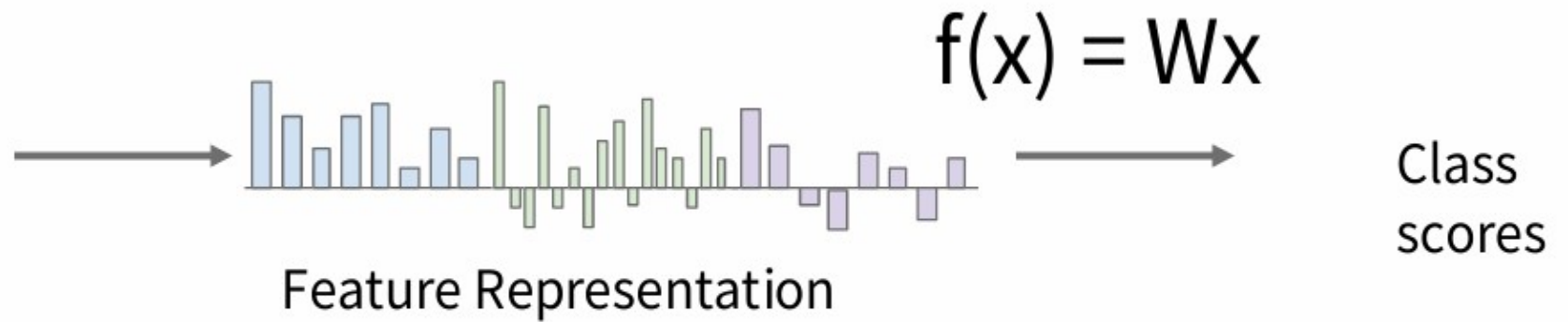
The spatial structure of images is destroyed!

Feature Extraction

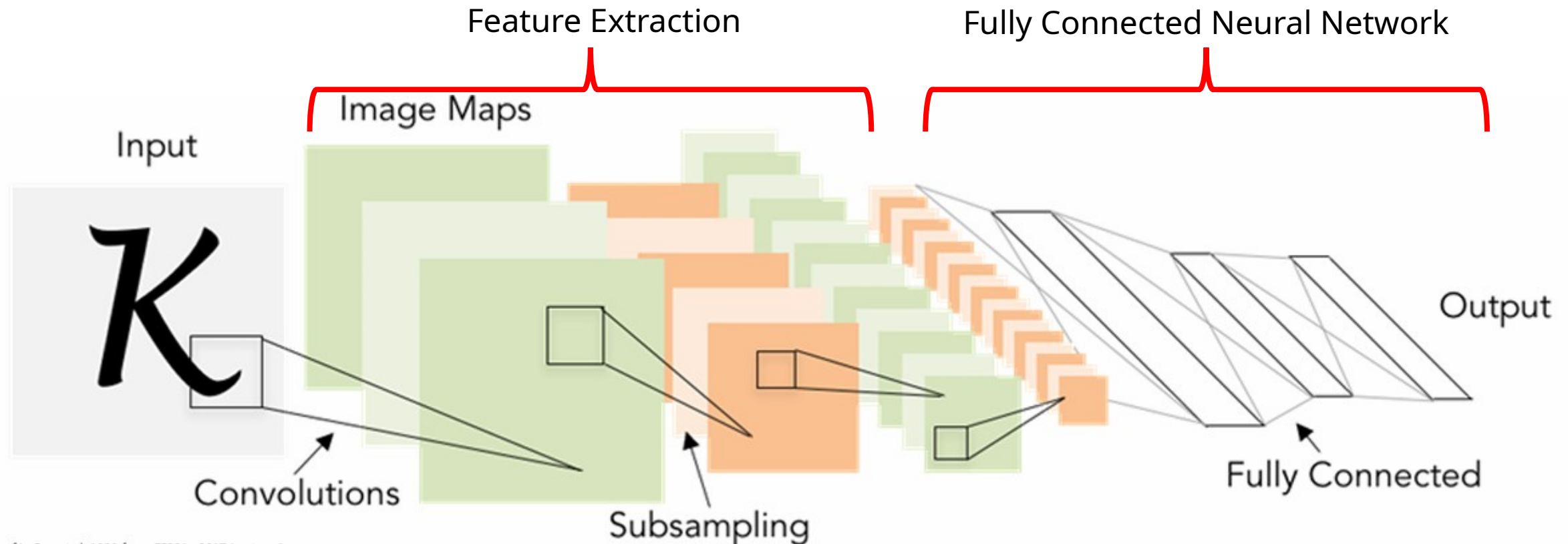


Feature Extraction

- Color histogram
- HOG
- Random Patches
- Feature engineering

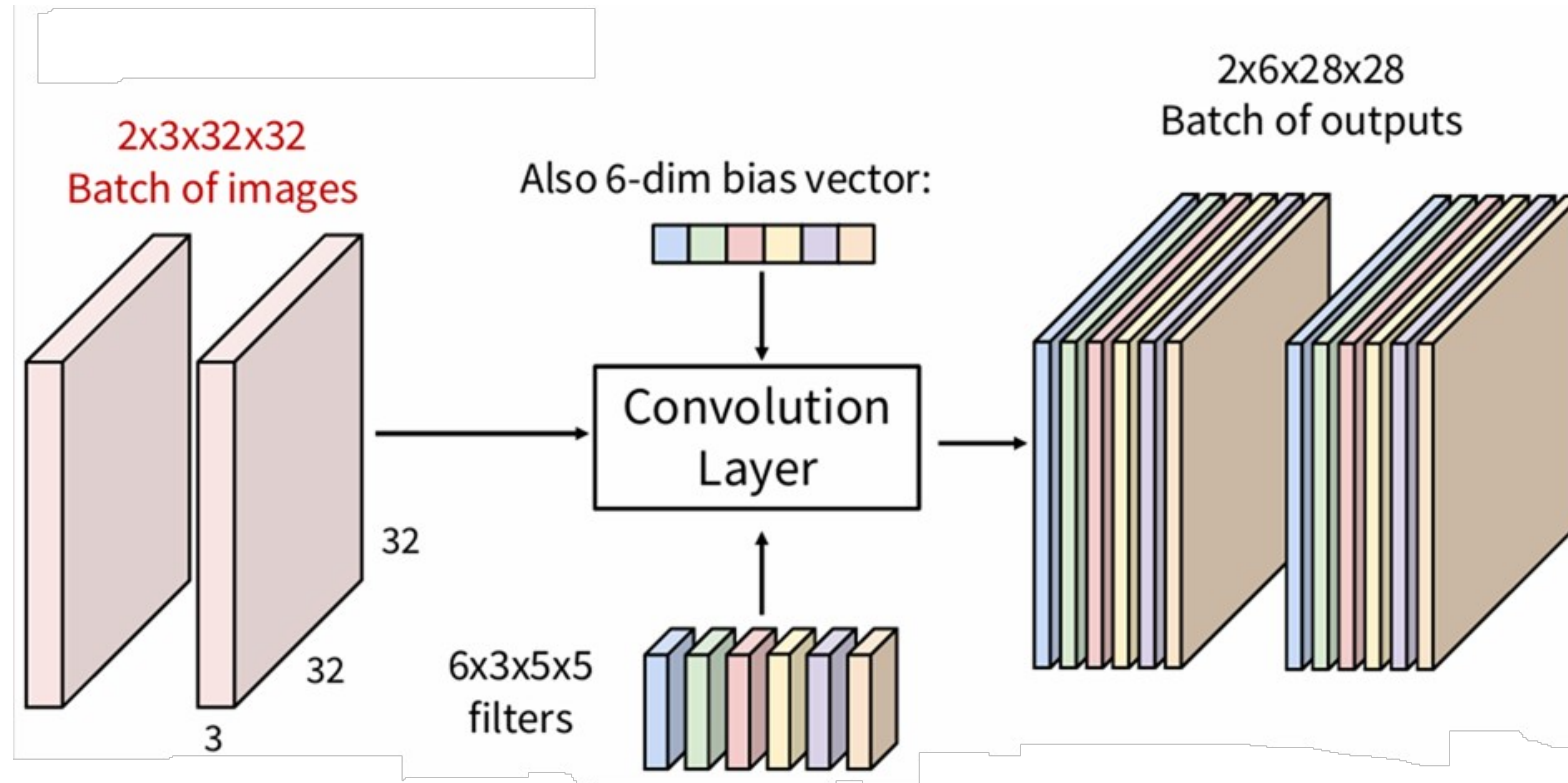


Convolutional Neural Networks



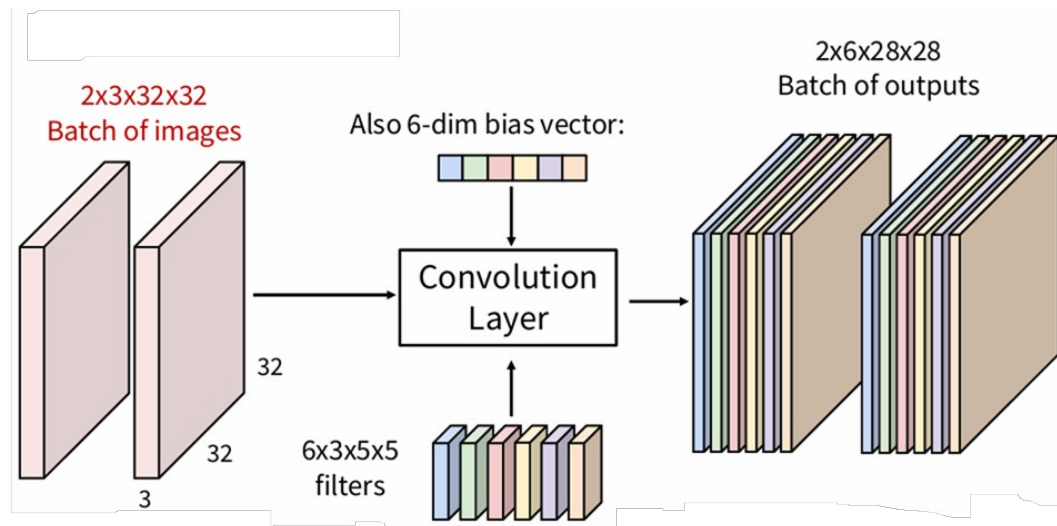
Convolutional Neural Networks

- Convolution Layer

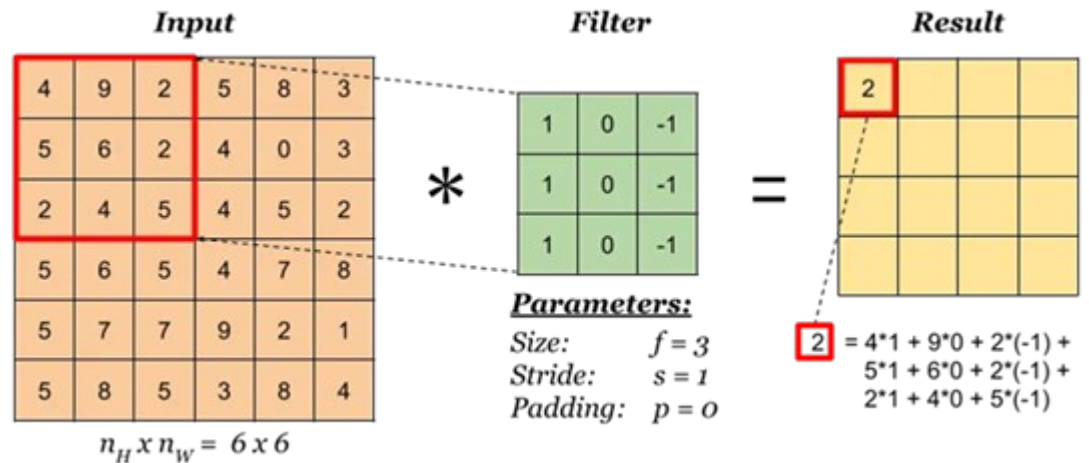


Convolutional Neural Networks

- Convolution Layer

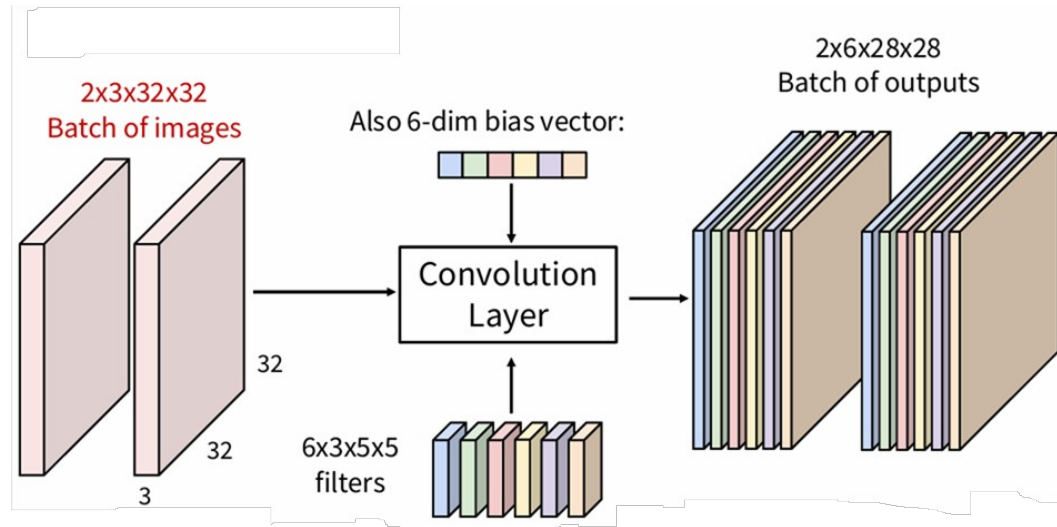


- A convolution operation



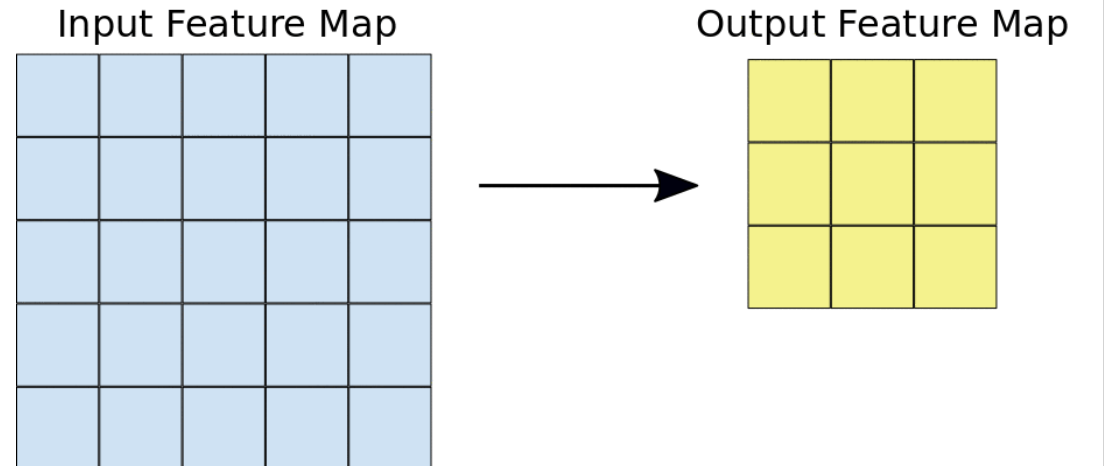
Convolutional Neural Networks

- Convolution Layer



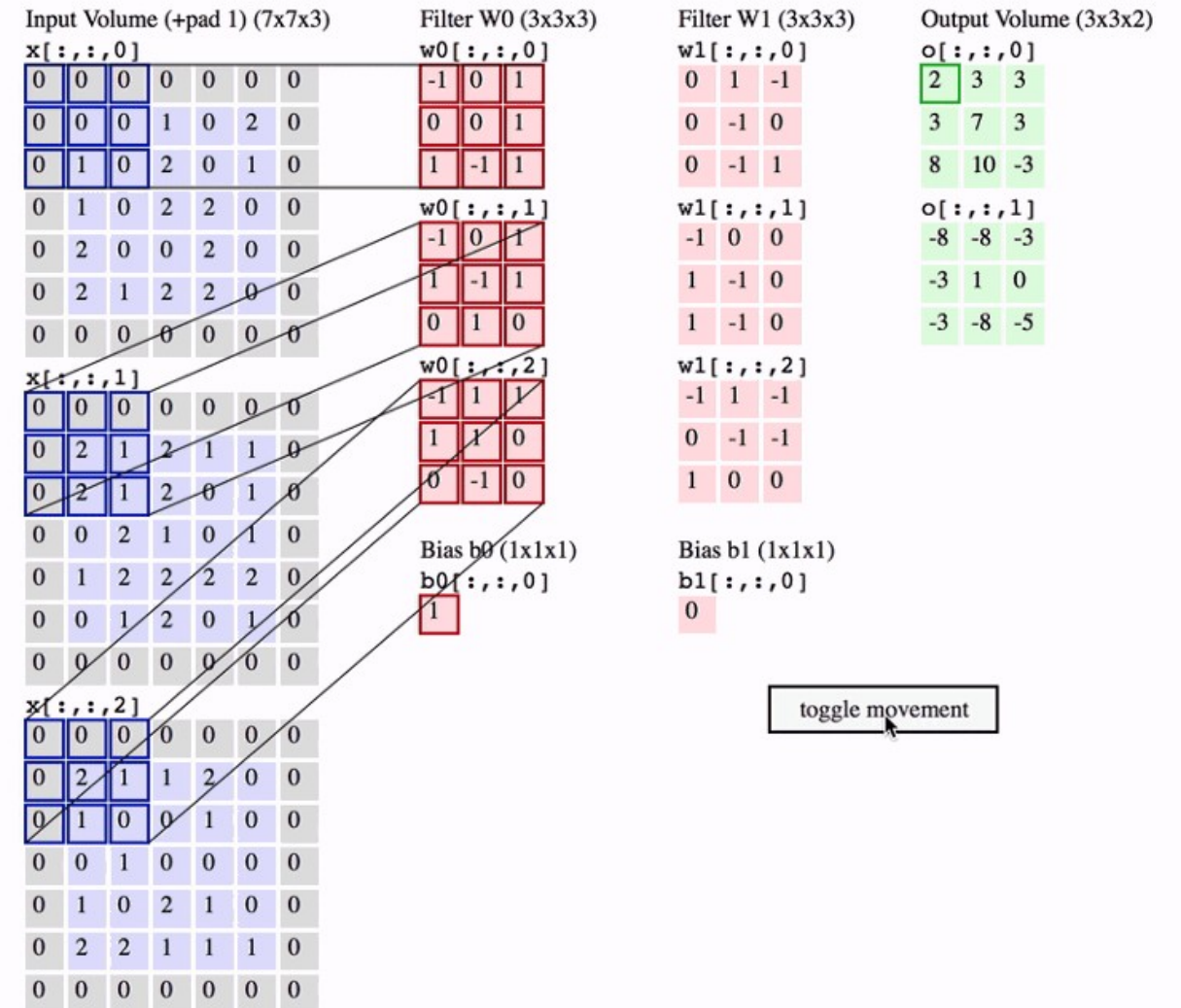
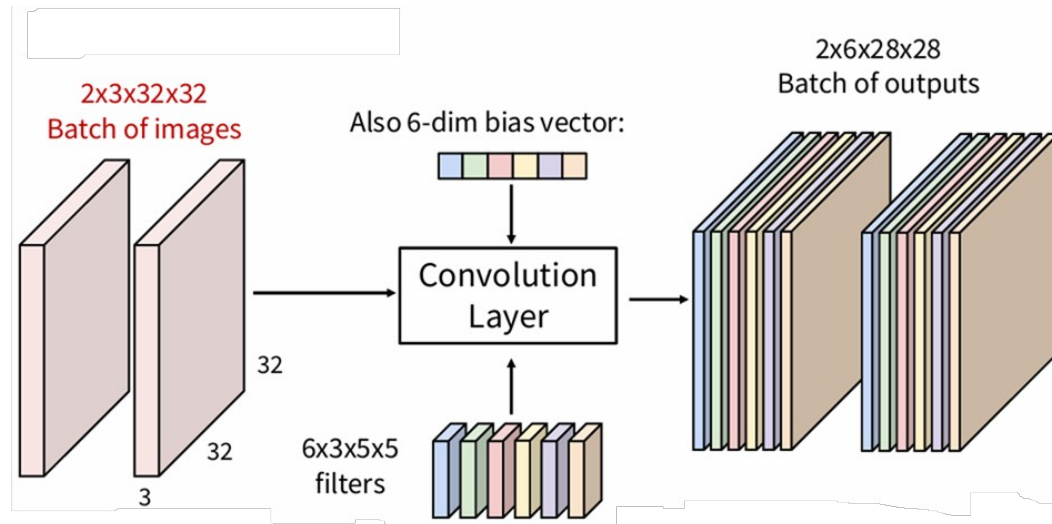
- Stride

- Here stride is 1



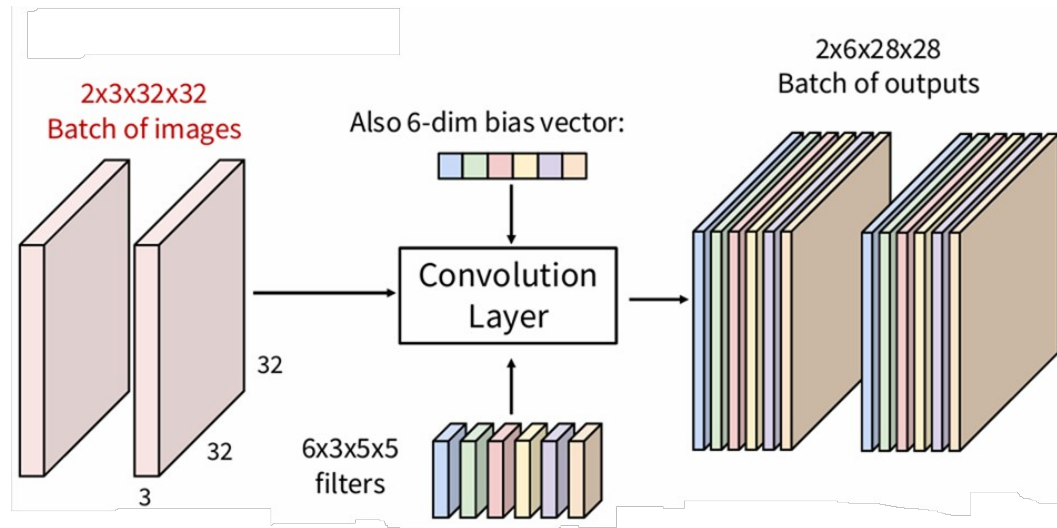
Convolutional Neural Networks

- Convolution Layer



Convolutional Neural Networks

- Convolution Layer

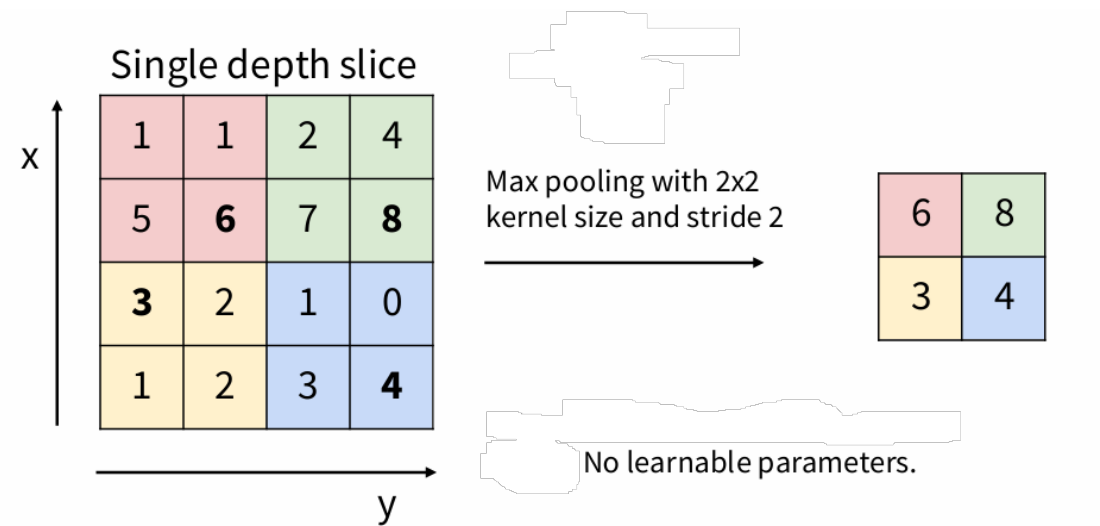


- Padding

- Avoid down-sampling with zero-padding.
- Input size W
- Filter size K
- Padding size P
- Output size $W - K + 1 + 2P$

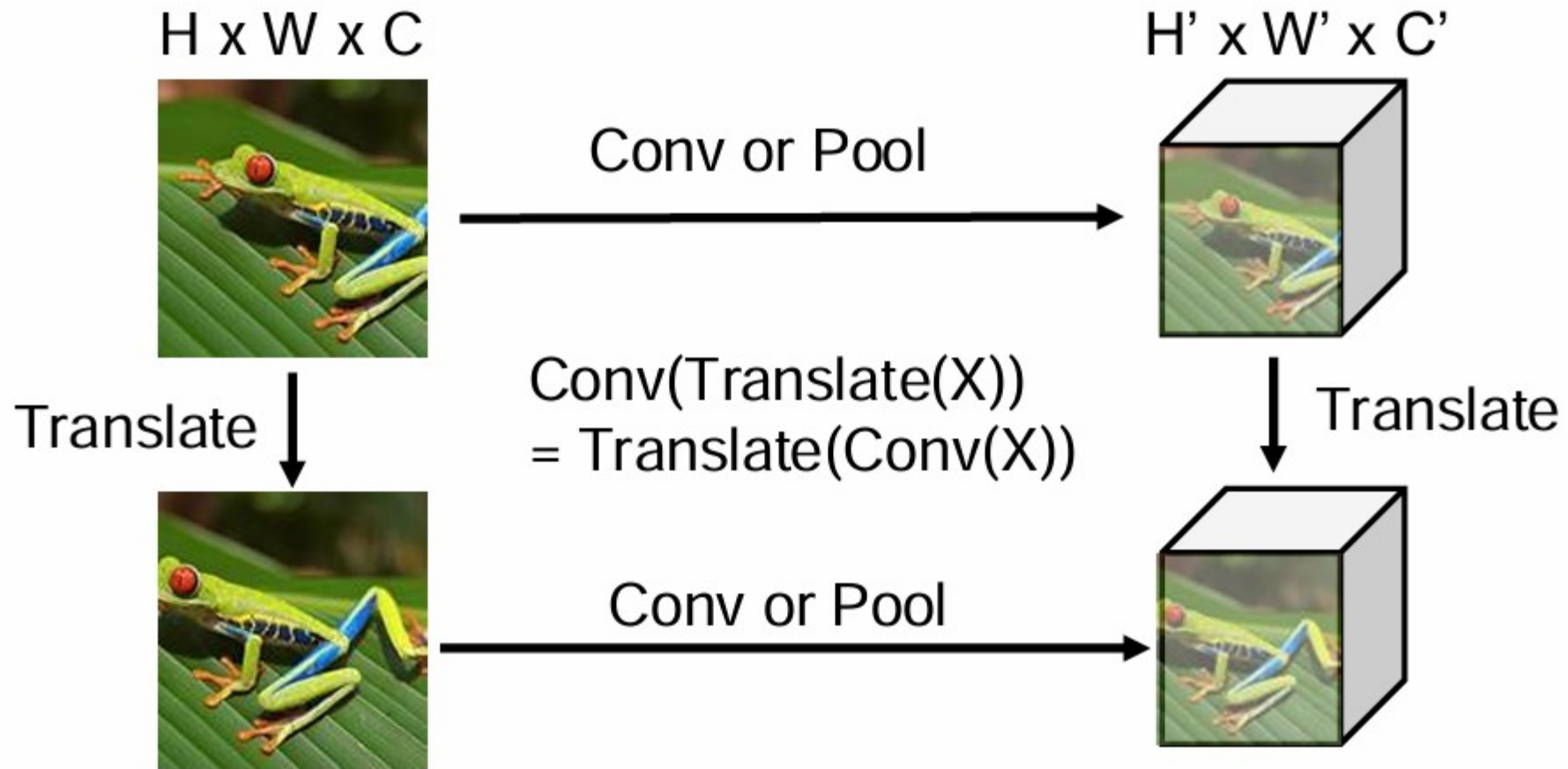
Convolutional Neural Networks

- Pooling
 - A way to downsample



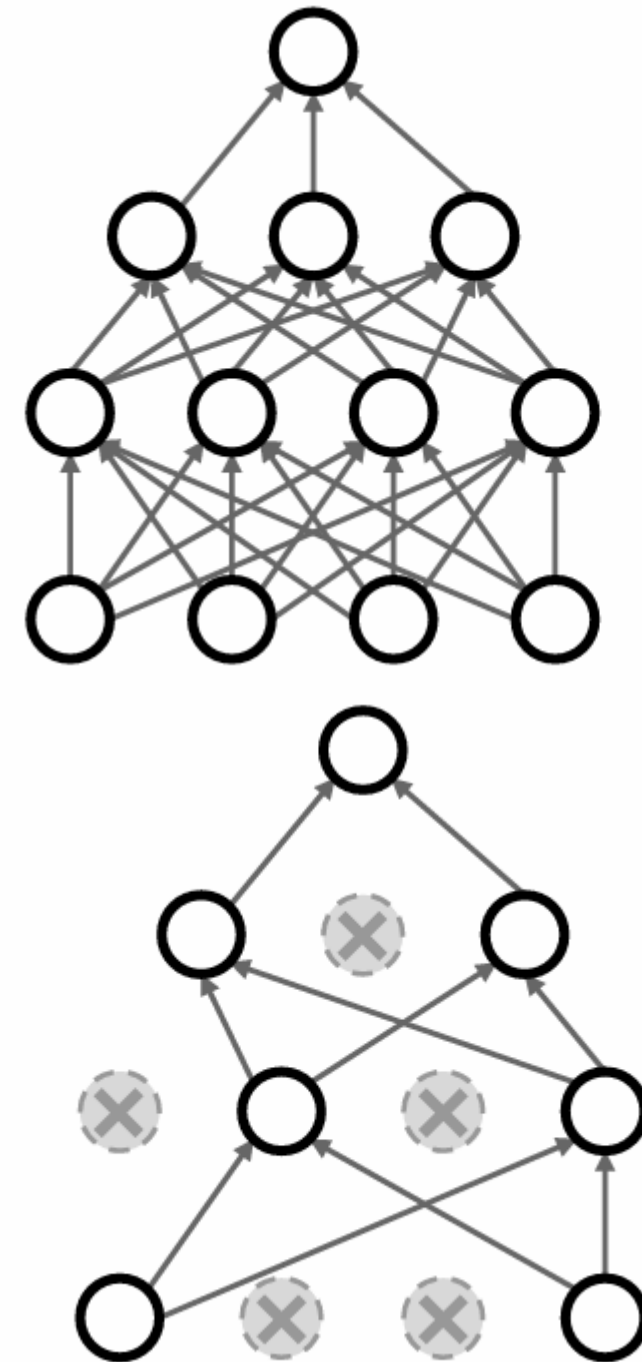
- No learnable parameters
- Max Pooling
- Average Pooling

Convolution and Pooling Translation Equivariance



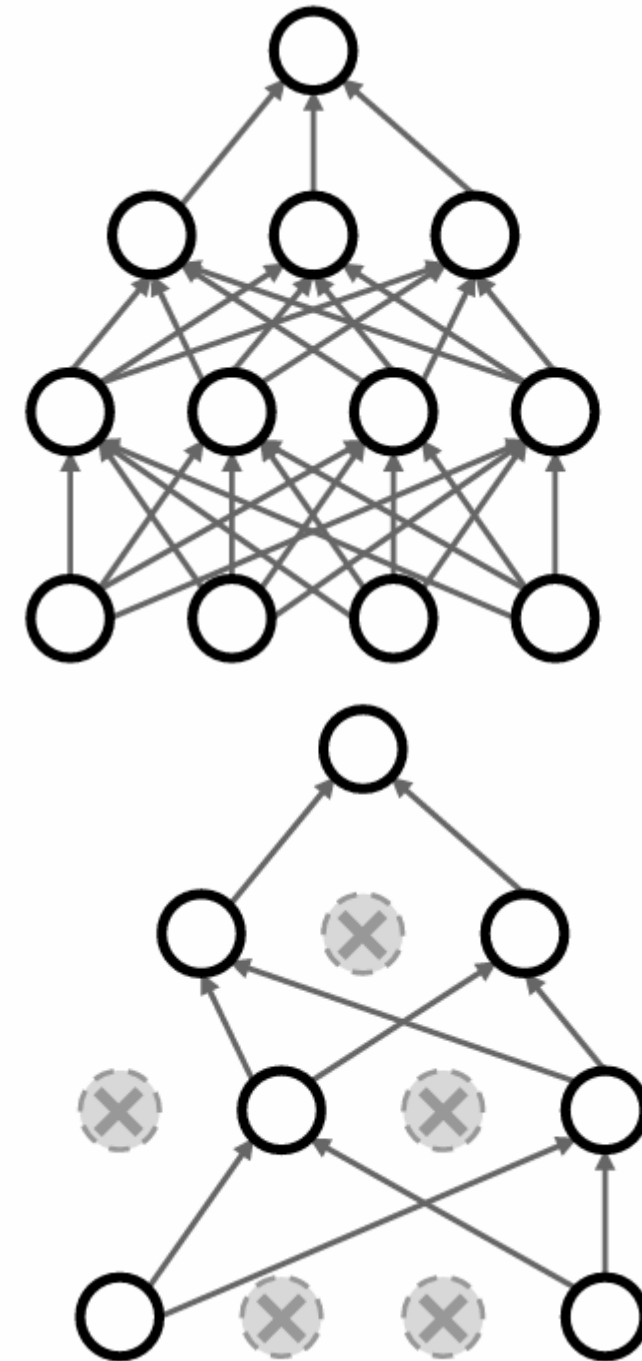
Dropout

- In each forward pass, randomly set some neurons to zero
- Probability of dropping is a hyperparameter; 0.5 is common



Dropout

- Why dropout is necessary?
 - As if training a large ensemble of models but sharing parameters
 - Each binary mask is one model
 - Dropout is used only at training time, no dropout at test time



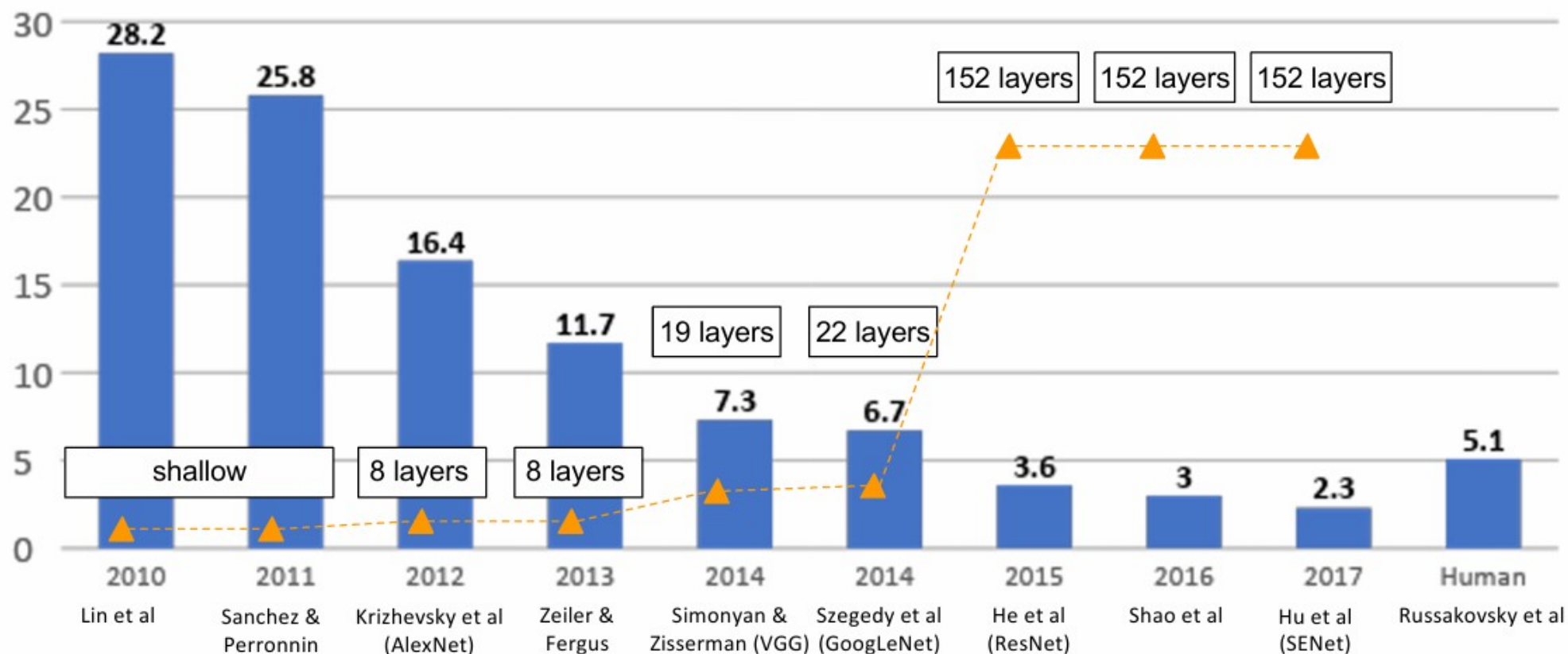
Batch Normalization

- z_i are values of z for each instance of the minibatch
- μ - mean
- σ – standard deviation
- ϵ – small constant
- γ and β are learned parameters

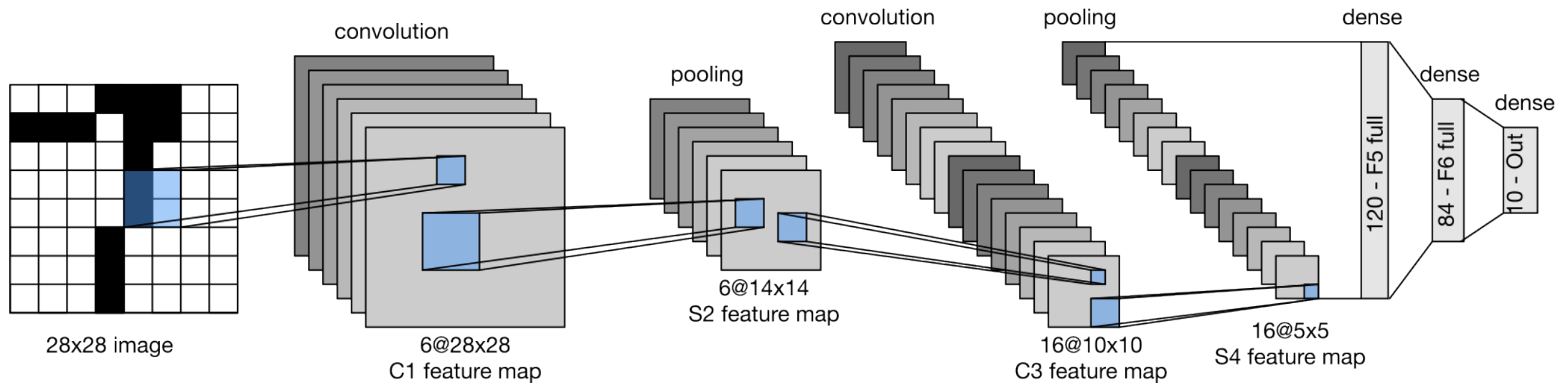
$$\hat{z}_i = \gamma \frac{z_i - \mu}{\sqrt{\epsilon + \sigma^2}} + \beta$$

CNN Architectures

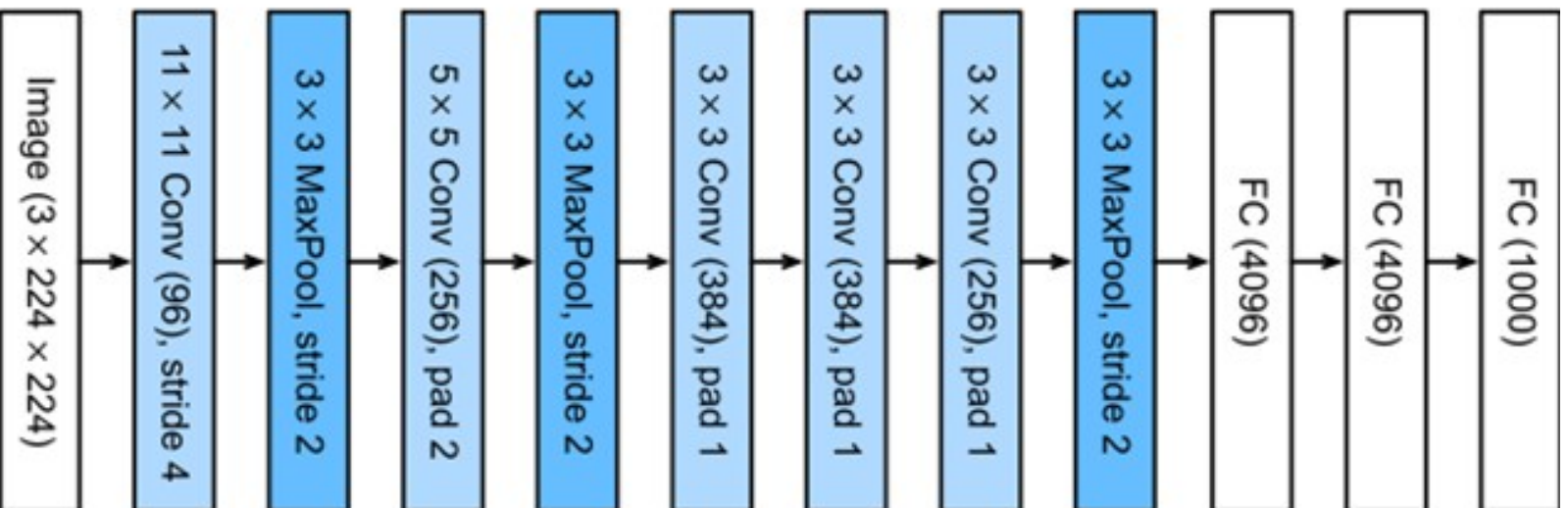
ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



Lenet

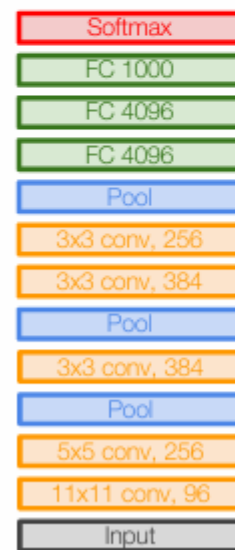


Alexnet



VGG

- Only uses 3*3 convolution layer
- Why?



AlexNet

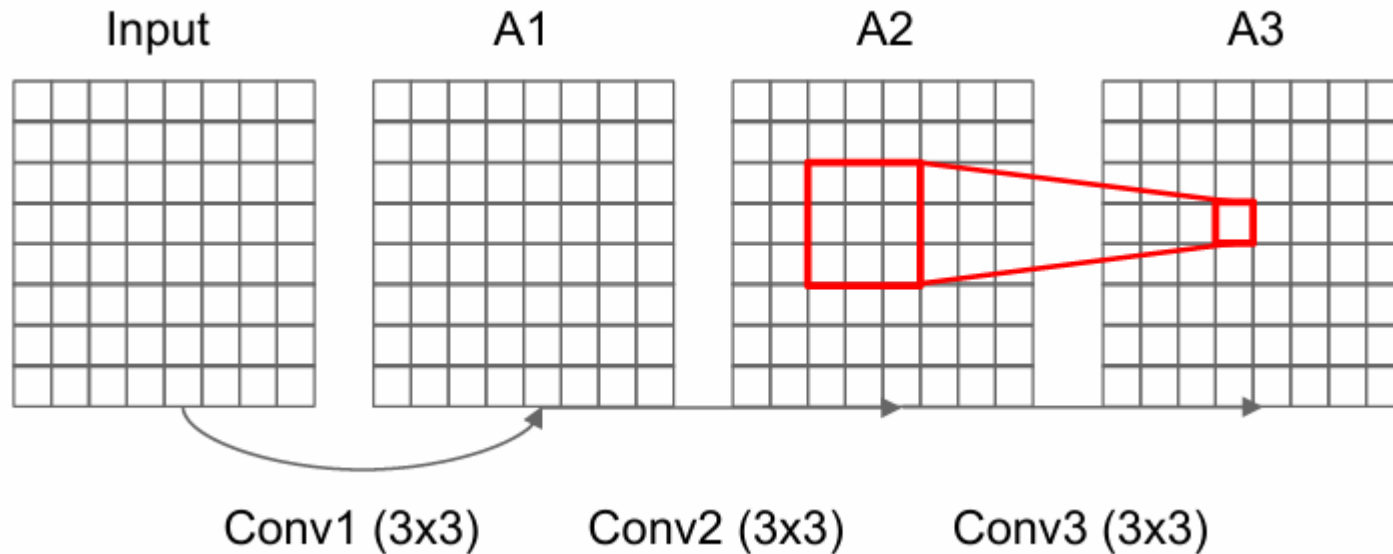


VGG16

VGG19

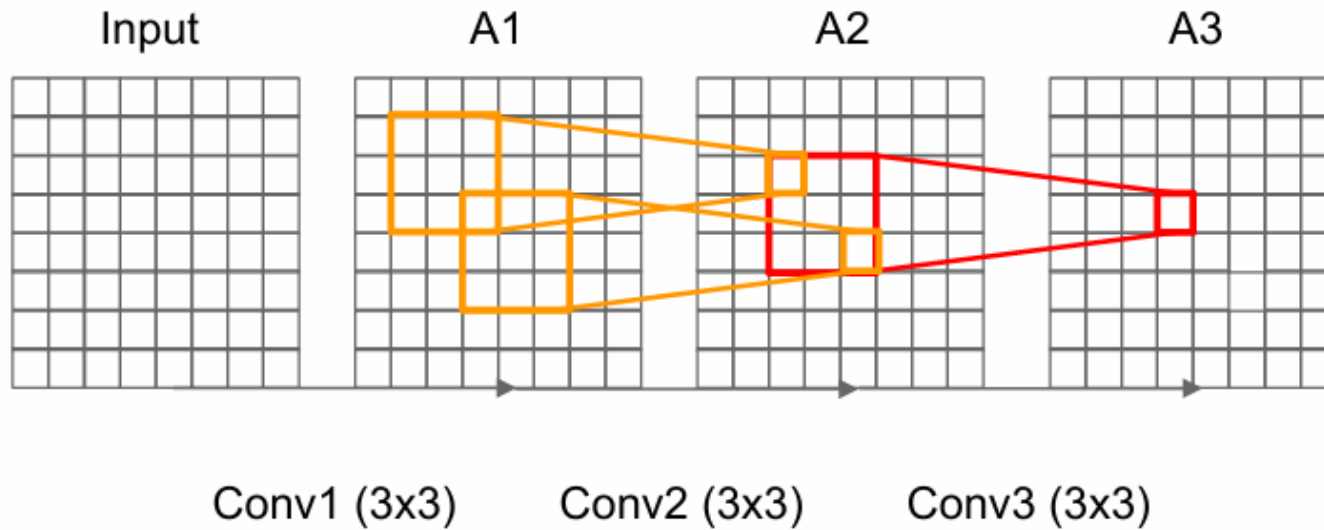
VGG

- Only uses 3*3 convolution layers
- Why?



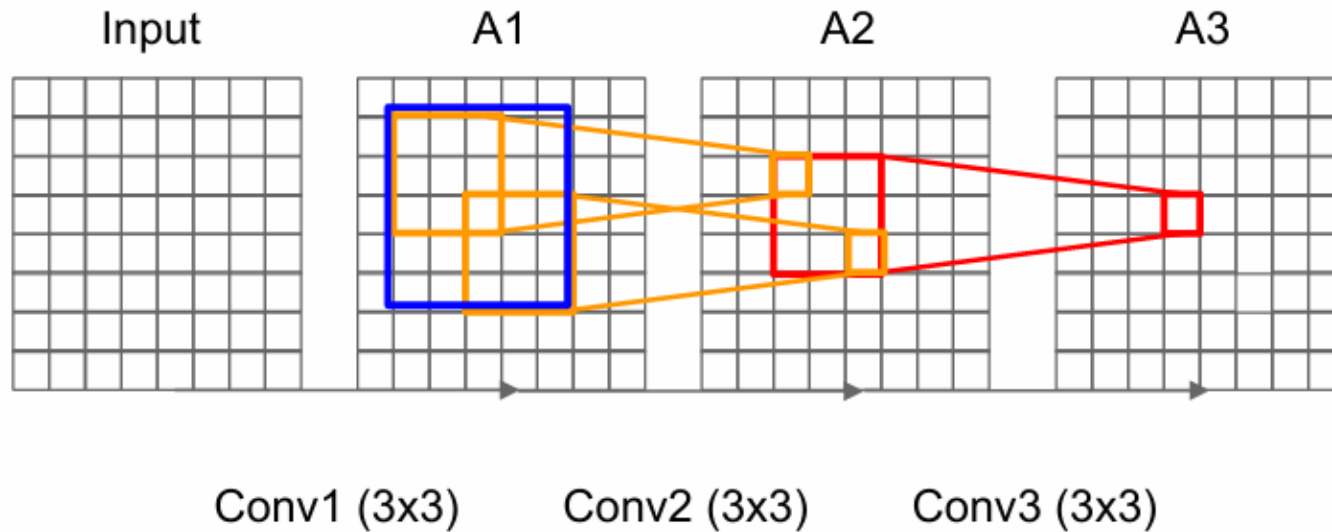
VGG

- Only uses 3*3 convolution layers
- Why?



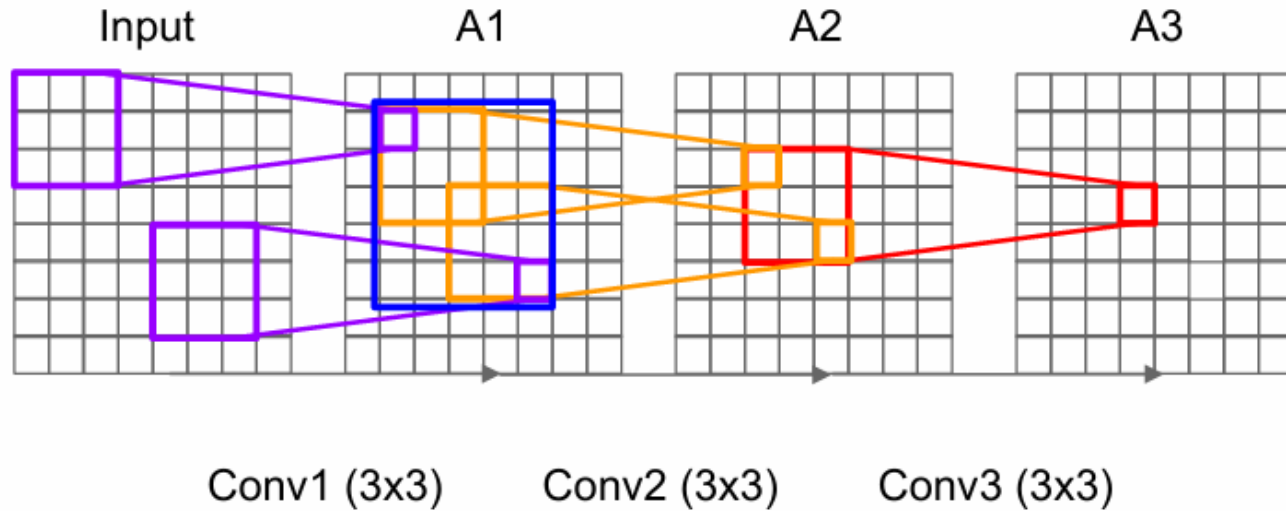
VGG

- Only uses 3*3 convolution layers
- Why?



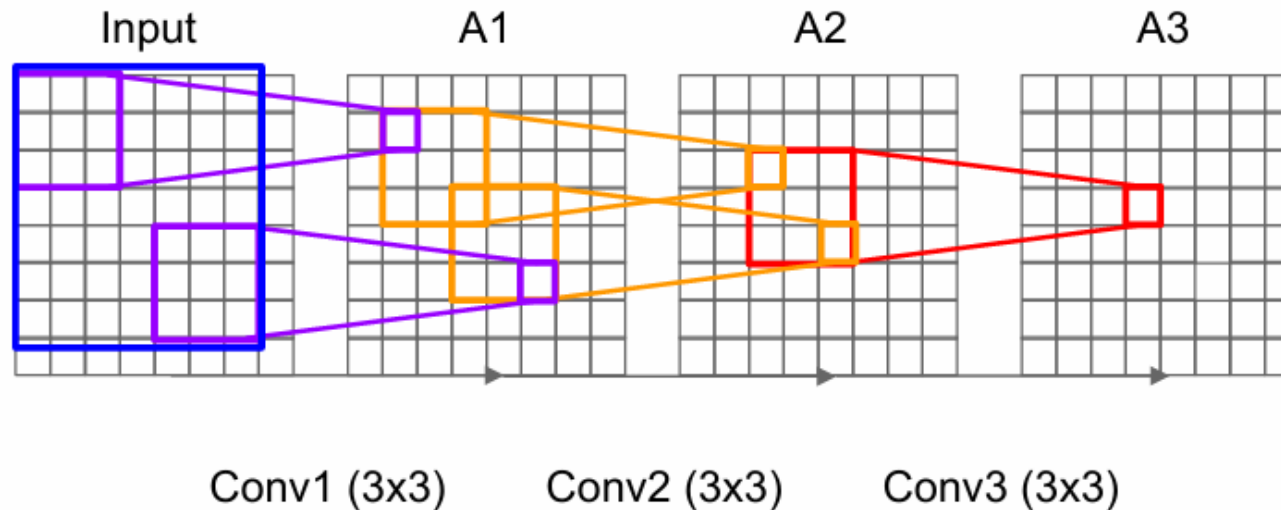
VGG

- Only uses 3*3 convolution layers
- Why?



VGG

- Only uses 3*3 convolution layers
- Why?

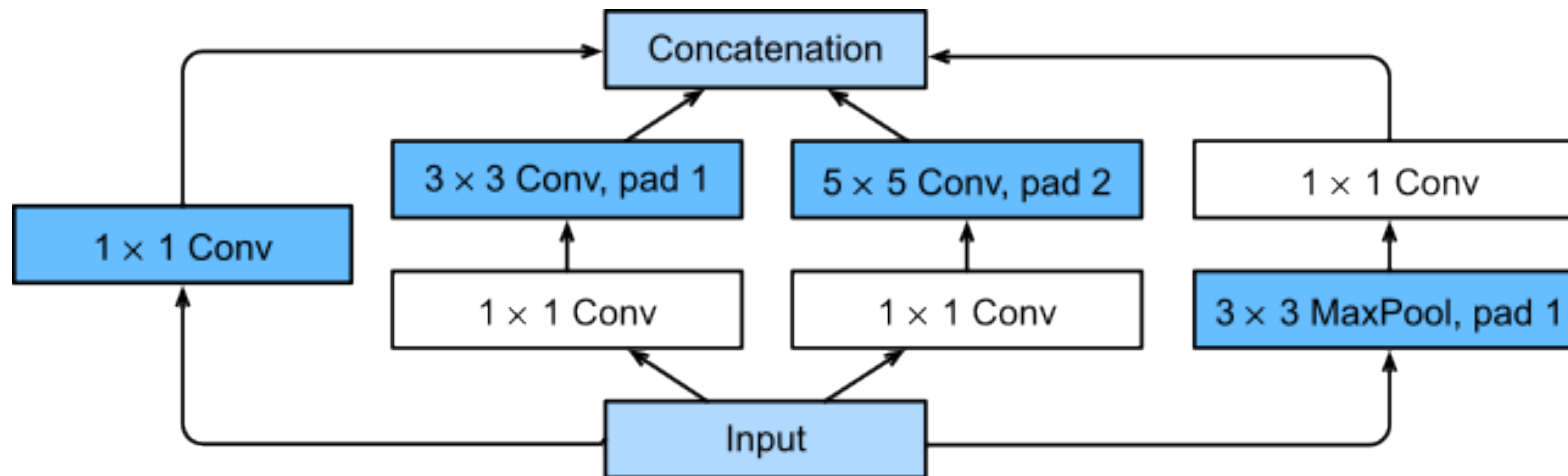


- Effective receptive field as one 7*7 conv layer



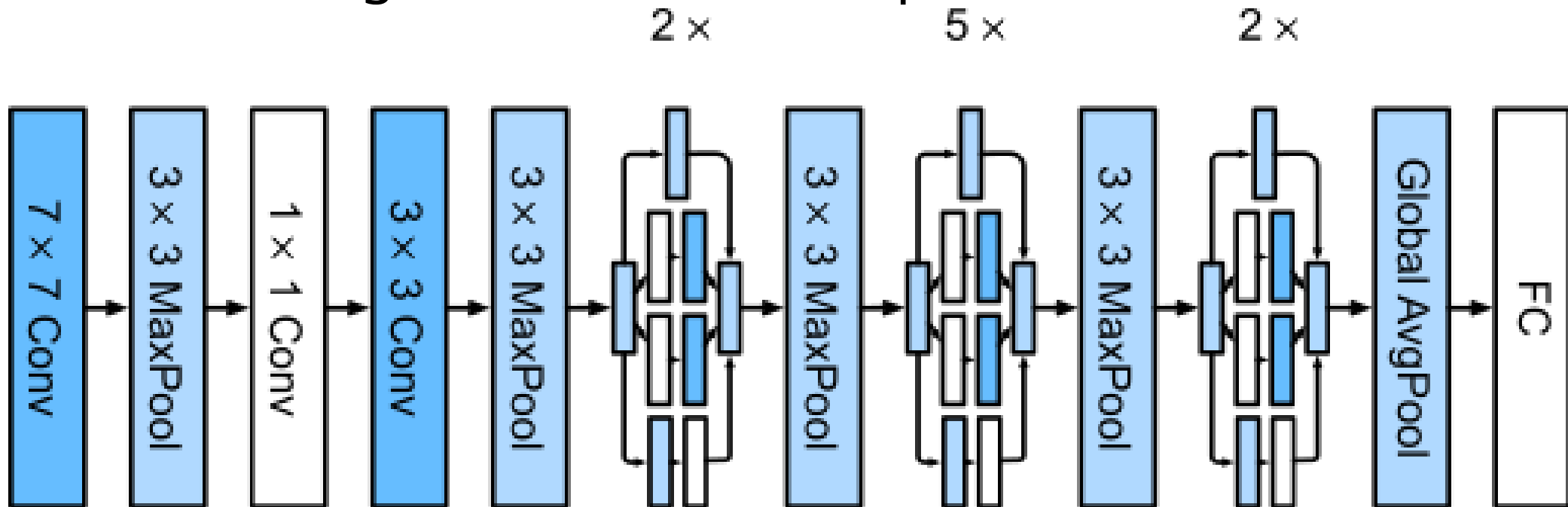
GoogLeNet

- Investigated which kernel size is the best
 - Employ a combination of various size kernels
- Basic building block is called inception block



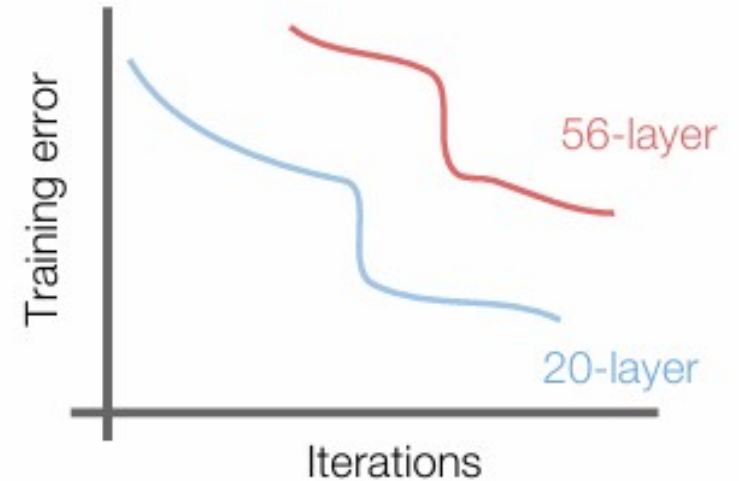
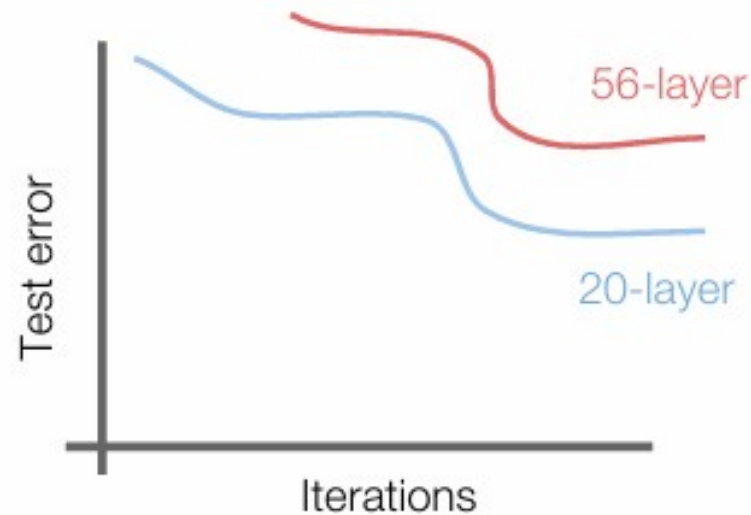
GoogLeNet

- Investigated which kernel size is the best
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Resnet

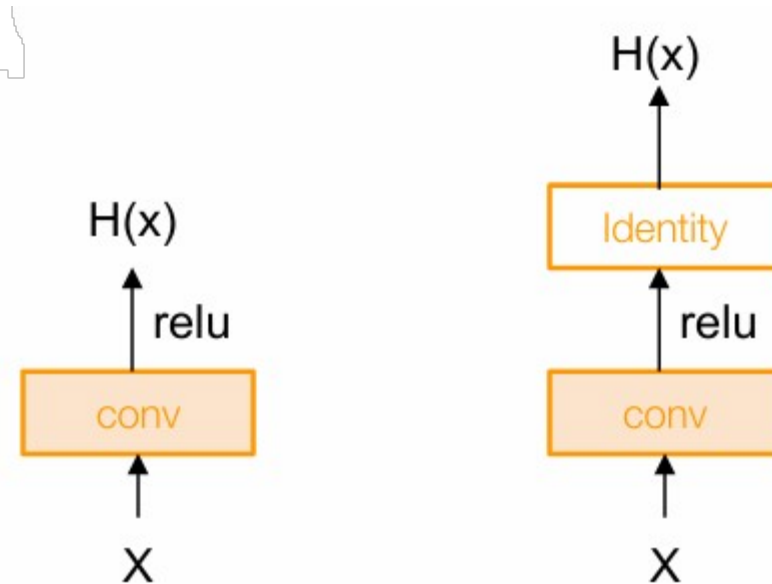
- Continue stacking layer to train deeper networks.



- 56-layer model performs worse on both test and training error.

Resnet

- The deeper model should learn to be at least as good as the shallower model
- Proof by Construction
 - Copying the learned layers from the shallower model and set additional layers to identity mapping.

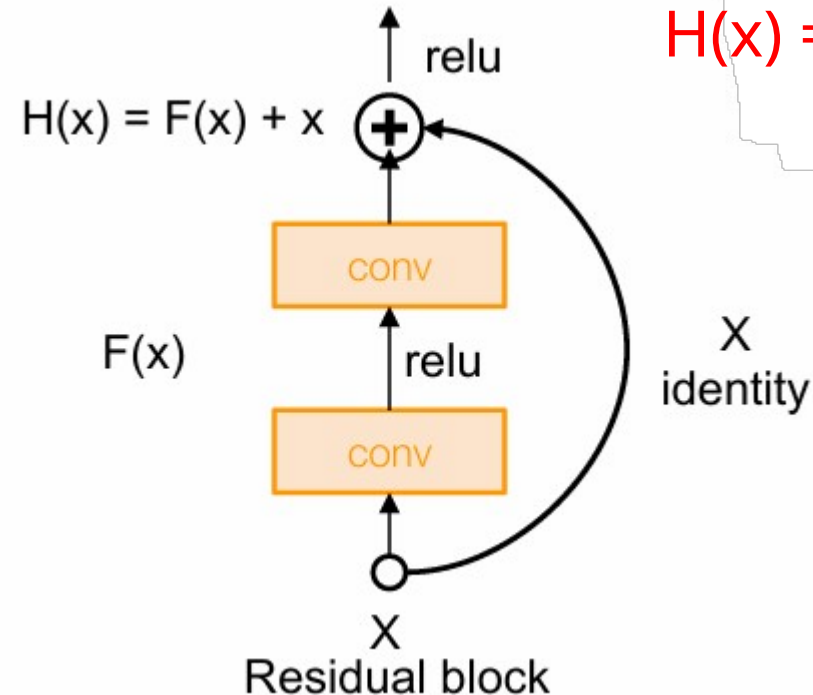
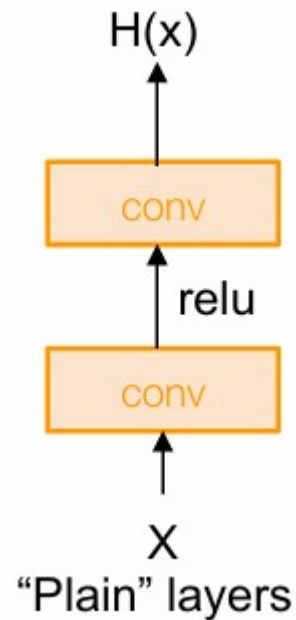


Resnet

- Deep models have more representation power (more parameters) than shallower models.
- Hypothesis: the problem is an optimization problem, deeper models are harder to optimize

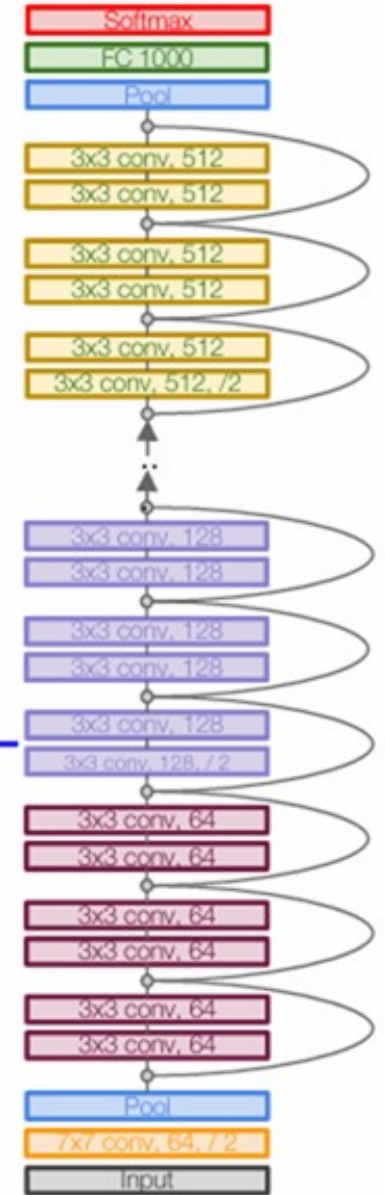
Resnet

- Use network layers to fit a residual mapping instead of directly trying to fit a desired underlying mapping



Identity Function,
 $H(x) = x$ if $F(x) = 0$

-
- Residual block
- X
- 3×3 conv
- relu
- 3×3 conv
- relu
- $F(x)$
- $+$
- X identity
- $F(x) + X$
- relu



Acknowledgement

- cs231n.stanford.edu